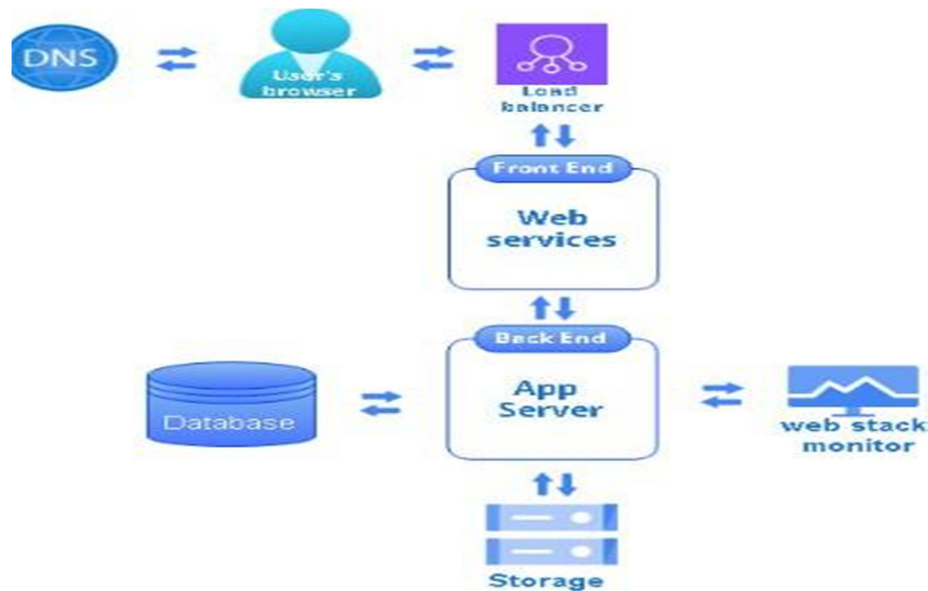


Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	27 June 2025
Team ID	LTVIP2025TMID53093
Project Name	HouseHunt : Finding your perfect rental Home
Maximum Marks	4 Marks

Technical Architecture – HouseHunt

HouseHunt is built using a **client-server architecture**, ensuring smooth interaction between users, agents, and admins. The system is divided into three main layers: **Frontend**, **Backend**, and **Database**. RESTful APIs connect the layers, enabling secure and efficient data exchange. Real-time chat and notifications are supported using **Socket.IO**.



Architecture Guidelines – HouseHunt

- The system includes core blocks:
 1. **Frontend:** React.js (Material UI, Bootstrap)
 2. **Backend:** Node.js + Express.js (REST APIs)
 3. **Database:** MongoDB (User, Complaint, Chat, Feedback data)
- **Infrastructure:**
 1. Local setup for development
 2. Cloud deployment via Vercel (frontend), Render or Railway (backend), MongoDB Atlas (database)
- **External Interfaces:**
 1. Gmail SMTP for emails
 2. Google OAuth for login
 3. Optional: Twilio for SMS
- **Data Storage:**
 1. All structured data in MongoDB
 2. Files/images stored via Firebase or AWS S3 (optional)
- **ML Model (Optional):**
 1. Future-ready for smart routing or auto-prioritization using ML

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How user interacts with the application (Web UI, etc.)	React.js, HTML, CSS, JavaScript, Material UI, Bootstrap
2	Backend	Server-side logic, API routes, authentication, CRUD operations	Node.js, Express.js
3	Authentication	Handles user login, registration, and JWT-based session contro	JSON Web Token (JWT)
4	API Testing	Testing API endpoints manually during development	Postman
5	Database	Data storage for users, complaints, chats, etc.	MongoDB (NoSQL)
6	State Management	Handling local component state and API response handling	React useState, useEffect, Axios
7	Image Upload	Uploading and saving property images to server	Multer (Node middleware)
8	Routing	Navigating between pages in frontend	React Router DOM
9	Admin Panel	Admin functionalities like owner approval	Custom-built React Components
10	Infrastructure	Hosting backend/frontend on cloud/local	Localhost, Render, Railway, Cloud Foundry, Kubernetes

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology Used
1	Open-Source Frameworks	Libraries and frameworks used	React.js, Express.js, Node.js, Mongoose, Socket.io
2	Security Implementations	Authentication, Authorization, Data Protection	JWT, bcrypt.js, CORS, HTTPS, SHA-256, Helmet
3	Scalable Architecture	Modular design for performance and growth	3-tier architecture, Microservices-ready, REST APIs
4	Availability	Ensures uptime, handles traffic	Load Balancers, Cloud Deployment, Clustered MongoDB
5	Performance	Optimized code for response time and user experience	Axios, CDN, Caching (Redis optional), Lazy Loading

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